

CSE587: Networks for AI/ML Systems

QUIZ 5 (30 mins) [Total: 13 points]

Nov 20, 2025

Your Roll No: _____ Name: _____

Q.1. Suppose we have two protocol options while designing a web service: TCP+TLS 1.3 and QUIC. **[2*3 = 6]**

- (a) You're building a mobile video streaming service where users frequently switch between Wi-Fi and 4G/5G. Which protocol do you pick, and how would you justify it to leadership?

Solution.

The appropriate choice is QUIC because:

- QUIC, because it completes a fresh connection in 1 RTT and can resume in 0 RTT. TCP+TLS 1.3 requires 3 RTTs (TCP handshake + 2 TLS handshakes).
- No HOL blocking means smoother streaming even when some packets are lost.
- Faster handshake helps in scenarios where sessions start/stop frequently.
- Short-lived API calls, mobile applications, and latency-sensitive workloads see large benefits from QUIC's faster handshakes.

- (b) You're designing a high-bandwidth data transfer service inside a controlled datacenter. There's no mobility, and middleboxes are under your control. QUIC or TCP+TLS 1.3?

Solution.

- Either can work, but TCP+TLS 1.3 may be simpler because:
- 3-RTT handshake is tolerable inside a DC with persistent connections.
- HOL blocking is less severe because controlled networks have low loss.
- Mature ecosystem and tooling for congestion control tuning.

QUIC doesn't add strong advantages unless you need multiplexing or rapid restarts.

- (c) How do the two protocols differ in terms of encryption? Does the encryption process result in network management challenges? Elaborate.

Solution.

TLS 1.3 encrypts payload data but leaves TCP headers visible to middleboxes. QUIC encrypts most of its transport metadata, including packet numbers and headers, exposing only what is necessary for routing via the connection ID. This eliminates middlebox interference but makes operations like traffic shaping harder. QUIC is preferred in environments where performance, privacy, and protocol evolution matter; TCP+TLS may fit legacy or enterprise network scenarios that rely on visible transport fields.

Q.2. What is the purpose of IPsec AH (Authentication Header) mode? In the IPsec AH mode, is the entire outer IP header included in the hash/digest computation? Why? **[1+1+1=3]**

Solution.

- AH mode: Provides integrity and authentication
- No, because some of the outer IP header fields, such as "TTL" change during transit, and such mutable fields should not be added.

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Q.3. What are the different challenges in providing secure communication/data transfer among multiple nodes (GPUs, CPUs, XPU) of AI/ML infrastructure? What are the new protocols/technologies being developed to address these challenges? **[2+2= 4]**

Solution.

- Reduction in throughput and increase in latency
 - Complex key management in huge infrastructure
 - Model/Data exposure due to data-in-use vulnerability, i.e., plaintext model/data in memory during computation and computation
 - Solutions: hardware-based Trusted Execution Environments (TEEs), Hardware-accelerated security primitives, Federated learning, Homomorphic encryption.
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